

Chapter 1

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Definition. Let V be a vector space. The **vectorspace complexification** of V is $V \otimes_{\mathbb{R}} \mathbb{C}$. However you can view $V \otimes_{\mathbb{R}} \mathbb{C}$ as $V \oplus iV$ with regular complex multiplication ie $(a + bi)(v_1 + iv_2) = (av_1 - bv_2) + i(bv_1 + av_2)$ ■

Definition. Let $\mathfrak{g}$ be a Lie Algebra with lie bracket $[\cdot, \cdot]$. $\forall x \in \mathfrak{g}$ let $\text{ad}_x := [y \rightarrow [x, y]]$. The **Killing Form** is a symmetric bilinear form $K : \mathfrak{g} \times \mathfrak{g} \rightarrow F$ where F is the field $\mathfrak{g}$ is over where $K(x, y) = \text{tr}(\text{ad}_x \circ \text{ad}_y)$ ■

Problem 1. Let V be a Lie algebra over $\mathbb{R}$

1. Let $V^{\mathbb{C}}$ be the vector space complexification $V^{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C}$. Show that the bracket operation on V extends consistently and uniquely to $V^{\mathbb{C}}$ so $V^{\mathbb{C}}$ becomes a complex lie algebra.
2. Let V be finite dimensional and let B and $B^{\mathbb{C}}$ be the killing forms of V and $V^{\mathbb{C}}$. Show that $B^{\mathbb{C}}|_{V \times V} = B$
3. Show that V semisimple iff $V^{\mathbb{C}}$ semisimple
- 4.

Proof. 1. $[v_1 + iv_2, w_1 + iw_2] = [v_1, w_1] + i[v_2, w_1] + i[v_1, w_2] - [v_2, w_2]$. This is bilinear and satisfies the jacobi identity.

2. $\text{ad}_x(\text{ad}_y(z_1 + iz_2)) = [x, [y, z_1 + iz_2]] = [x, [y, z_1] + i[y, z_2]] = [x, [y, z_1]] + i[x, [y, z_2]]$. The matrix of ad_x, ad_y is the same in V and $V^{\mathbb{C}}$. thus B and $B^{\mathbb{C}}$ must be the same restricted to $V \times V$.

3. $\implies$: Suppose V is semisimple. Assume the contrary that $V^{\mathbb{C}}$ is not semisimple, thus $\exists x \in V^{\mathbb{C}}$ such that $\forall y \in V^{\mathbb{C}}, B_{V^{\mathbb{C}}}(x, y) = 0$. $x = x_1 + ix_2$ for $x_1, x_2 \in V$. Thus $B_{V^{\mathbb{C}}}(x, y) = B_{V^{\mathbb{C}}}(x_1 + ix_2, y) = B_{V^{\mathbb{C}}}(x_1, y) + iB_{V^{\mathbb{C}}}(x_2, y)$. Note $\forall y \in V, B_{V^{\mathbb{C}}}(x_i, y) = B_V(x_i, y) \in \mathbb{R}$. Thus if $B_V(x_1, y) + iB_V(x_2, y) = 0 \implies B_V(x_1, y) = 0$ and $B_V(x_2, y) = 0$. Thus B_V is degenerate which is a contradiction since V is semisimple.

$\impliedby$: Suppose $V^{\mathbb{C}}$ is semisimple. Assume the contrary, V is not semisimple, then B_V is degenerate. Thus $\exists x \in V \ni \forall y \in V, B_V(x, y) = 0$. $\forall y = y_1 + iy_2 \in V^{\mathbb{C}}, B_{V^{\mathbb{C}}}(x, y) = B_{V^{\mathbb{C}}}(x, y_1) + iB_{V^{\mathbb{C}}}(x, y_2) = B_V(x, y_1) + iB_V(x, y_2) = 0$. Thus $B_{V^{\mathbb{C}}}$ is degenerate. Thus by contradiction, V must be semisimple.

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